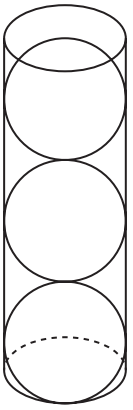
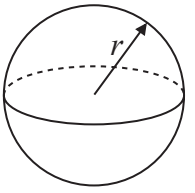
8 A hollow cylinder has radius r cm and height $6r$ cm.
3 spheres, also of radius r cm, are put into the cylinder.



Volume of sphere $= \frac{4}{3}\pi r^3$



(a) Work out the proportion of the cylinder that is **not** filled by the spheres.

Cylinder: $\pi r^2 \times 6r = 6\pi r^3$.

1 mark
Cylinder volume

Three spheres: $3 \times (4/3)\pi r^3 = 4\pi r^3$.

1 mark
Sphere total ÷ cylinder

Empty volume $= 6\pi r^3 - 4\pi r^3 = 2\pi r^3$.

Empty fraction $= \frac{2}{6} = \frac{1}{3}$

1 mark
Empty fraction1/3

(3)

The height of the cylinder is increased by $2r$ cm.
Another sphere of radius r cm is put into the cylinder.

Malcolm says,

"There is no change in the proportion of the cylinder **not** filled by the spheres."

(b) Is Malcolm correct?
Justify your answer.

Four spheres: filled fraction = (16/3) ÷ 8 = 2/3.

Yes: the empty fraction remains 1/3.

1 mark
Same proportion

(1)

(Total for Question 8 is 4 marks)